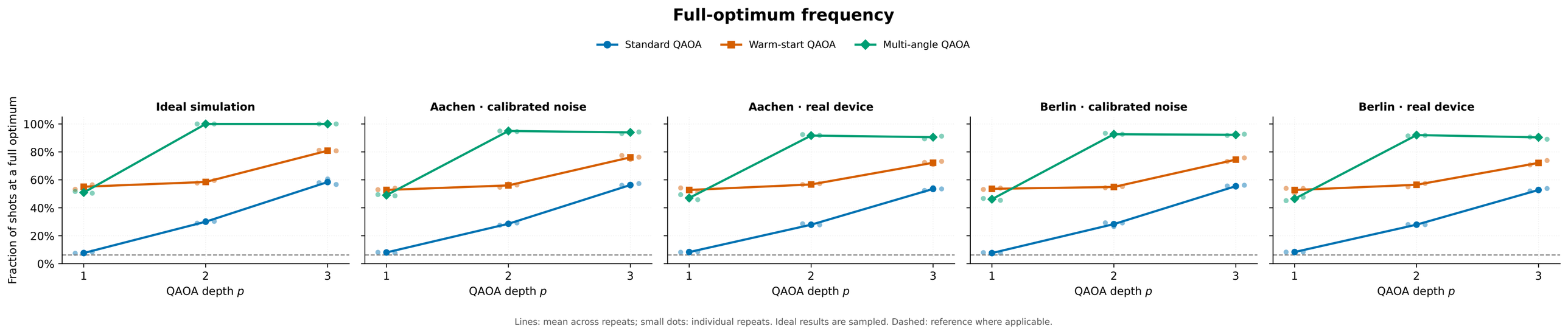

Full-optimum frequency
Standard QAOA
Warm-start QAOA
Multi-angle QAOA
Ideal simulation
Aachen · calibrated noise
Aachen · real device
Berlin · calibrated noise
Berlin · real device
Fraction of shots at a full optimum
100%
80%
60%
40%
20%
0%
1
2
3
QAOA depth p
Lines: mean across repeats; small dots: individual repeats. Ideal results are sampled. Dashed: reference where applicable.